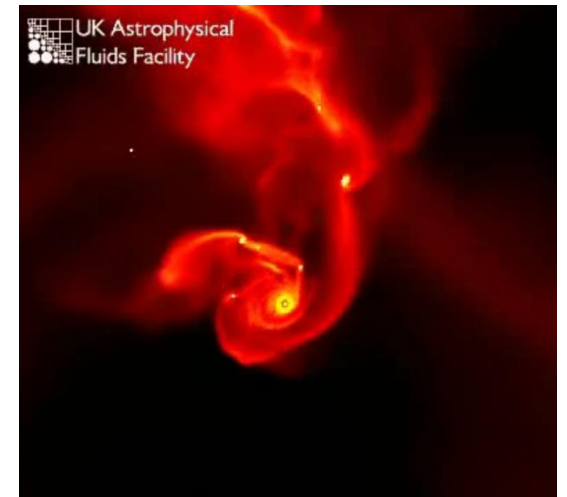
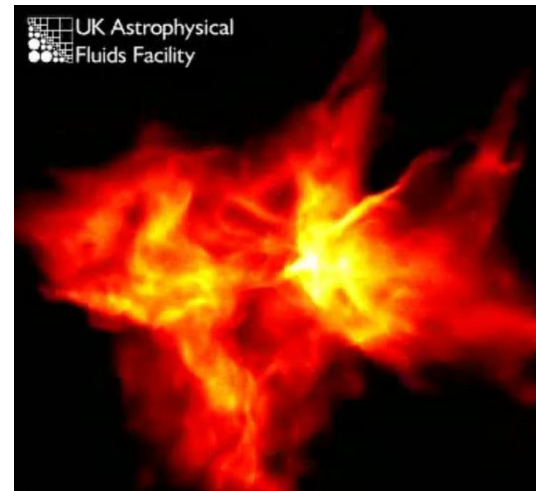
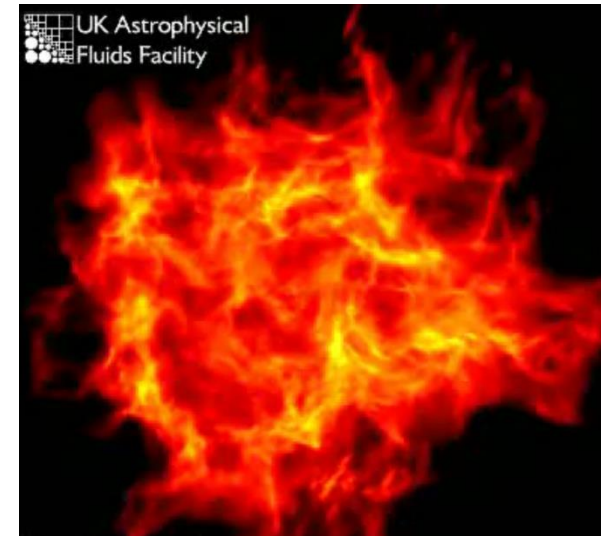
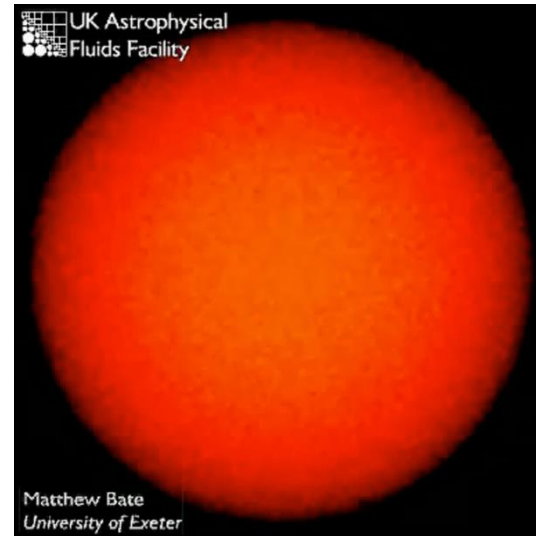


Jeans Instability

Jeans instability is responsible for the generation of structure in selfgravitating media.

The simulation on the right (Bate 2003) shows the fragmentatin of a 50 solar mass molecular cloud into a star cluster.

The process is driven by gravity, leading to collapse, and pressure, resisting collapse.



Jeans Instability

Sound waves removes compressibility.

To collapse, a region must then contract gravitationally faster than acoustics can remove the compression.

Timescale for acoustics: sound-crossing time

$$t_c = \lambda / c_s$$

Timescale for gravity: free-fall time

$$t_g = 1 / (G\rho)^{1/2}$$

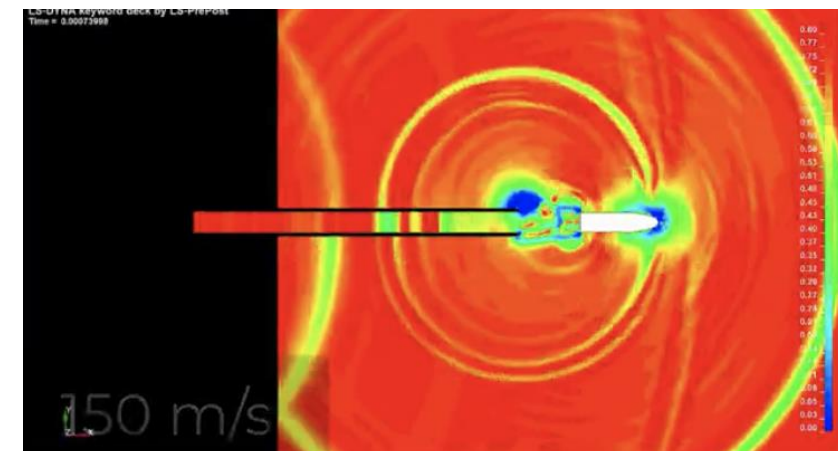
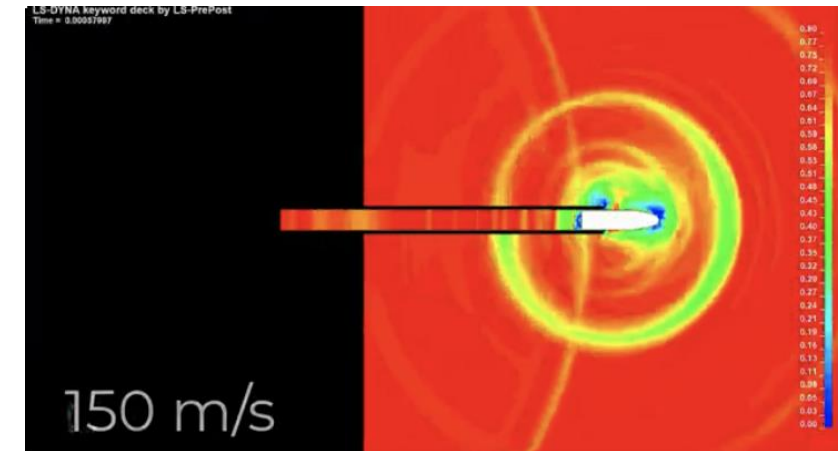
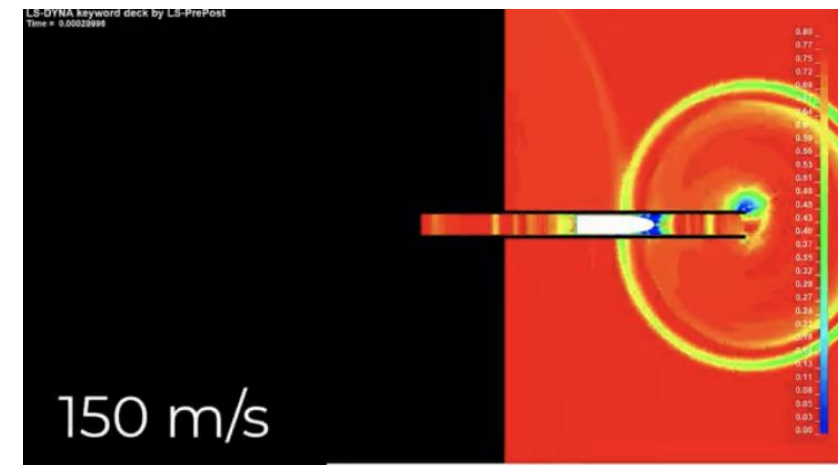
Comparing the scales, a region of length R will collapse if t_g is shorter than t_s

$$\lambda > c_s / (G\rho)^{1/2}$$

This is the *Jeans length*. It encloses a mass $\rho(\lambda/2)^3$, the *Jeans mass*

$$M_J \sim c_s^3 / (8G^{3/2} \rho^{1/2})$$

This analysis is off from the actual λ from rigorous linear analysis by a factor π^{-1} .



Hydro equations

$$\begin{aligned}\frac{\partial \rho}{\partial t} + (\mathbf{u} \cdot \nabla) \rho &= -\rho \nabla \cdot \mathbf{u} \\ \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} &= -\frac{1}{\rho} \nabla p - \nabla \Phi \\ \nabla^2 \Phi &= 4\pi G \rho\end{aligned}$$

Linearization

$$\begin{aligned}\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \mathbf{u}' &= 0 \\ \frac{\partial \mathbf{u}'}{\partial t} + \frac{c_s^2}{\rho_0} \nabla \rho' + \nabla \Phi' &= 0 \\ \nabla^2 \Phi' - 4\pi G \rho' &= 0\end{aligned}$$

Fourier modes

$$\begin{aligned}-\omega \hat{\rho} + \rho_0 k \hat{u} &= 0 \\ -\omega \hat{u} + \frac{c_s^2}{\rho_0} k \hat{\rho} + k \hat{\Phi} &= 0 \\ k^2 \hat{\Phi} + 4\pi G \hat{\rho} &= 0\end{aligned}$$

Jeans Instability

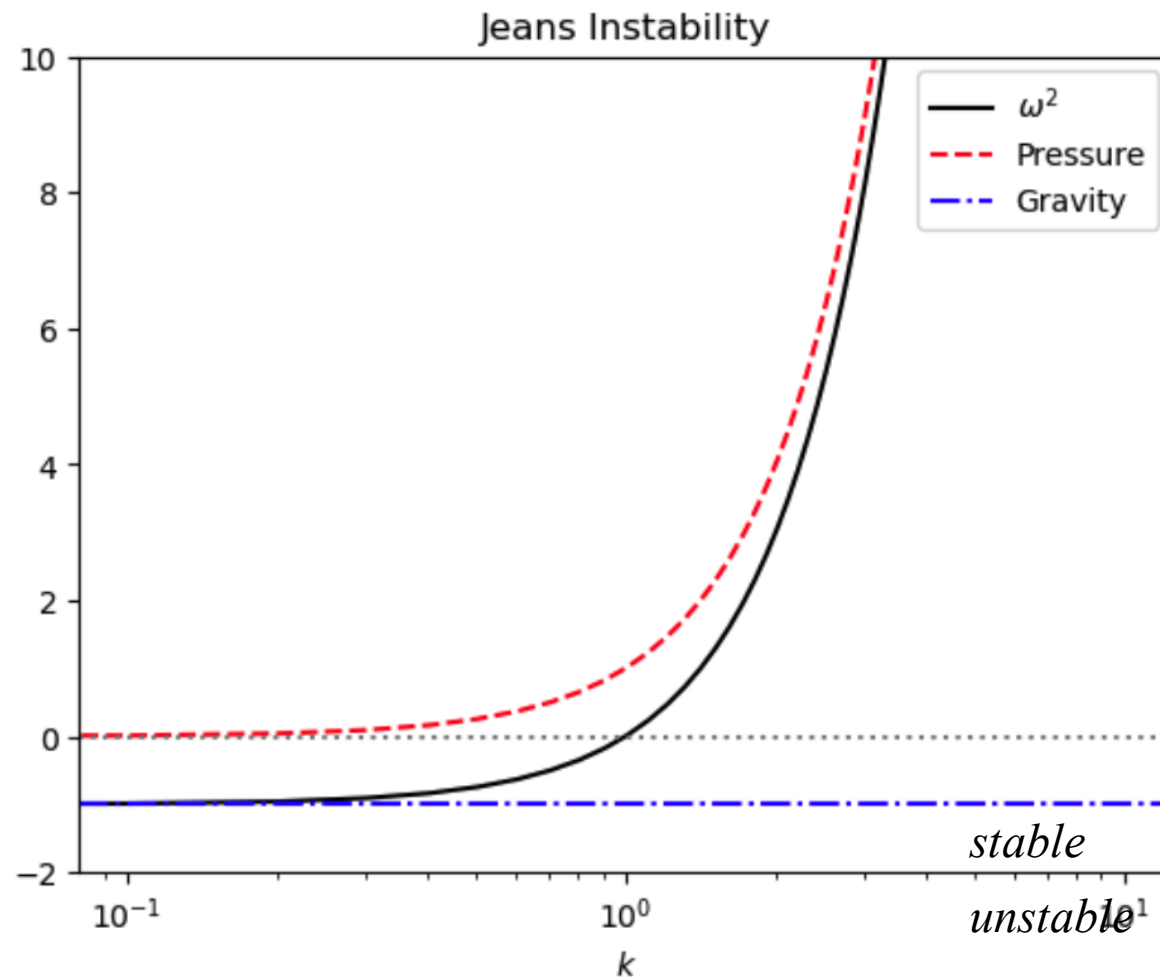
Dispersion relation

$$\omega^2 = k^2 c_s^2 - 4\pi G \rho_0$$

$$k_J^2 \equiv \frac{4\pi G \rho_0}{c_s^2} = \frac{4\pi G \rho_0 m_p \mu}{k_B T}$$

$$\lambda_J \equiv \frac{2\pi}{k_J} = c_s \left(\frac{\pi}{G \rho_0} \right)^{1/2} = \left(\frac{\pi k_B T}{G \rho_0 m_p \mu} \right)^{1/2}$$

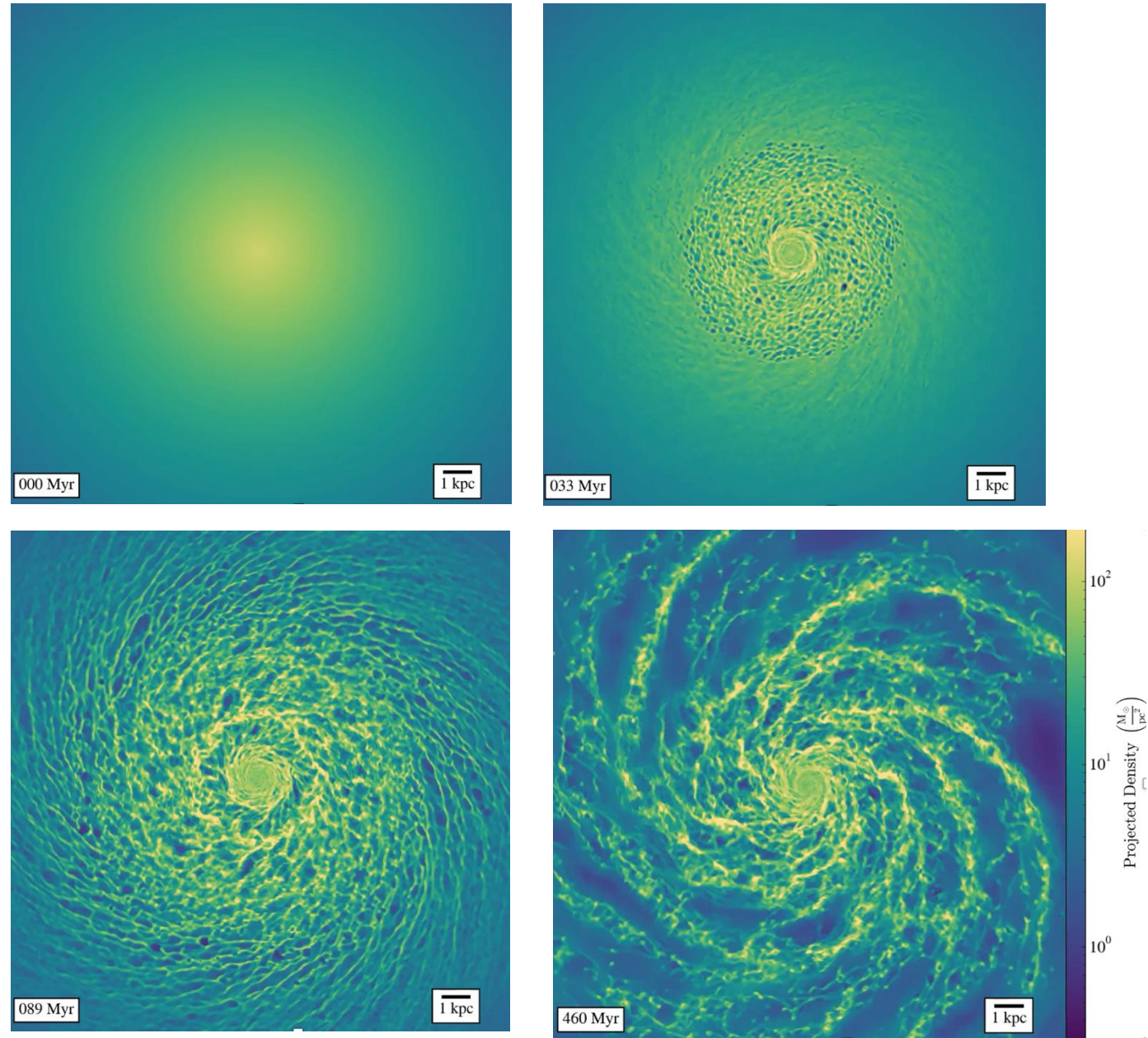
$$M_J \equiv \left(\frac{\lambda_J}{2} \right)^3 \rho_0 = \left(\frac{\pi k_B T}{4 G m_p \mu} \right)^{3/2} \frac{1}{\rho_0^{1/2}}$$



Toomre Instability

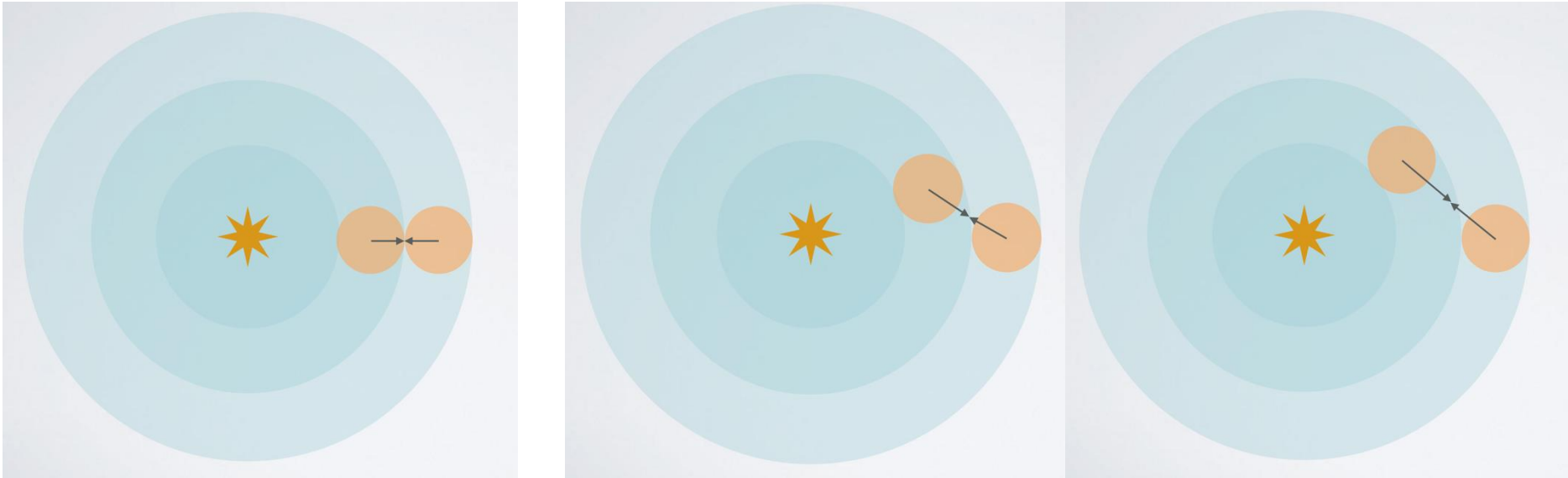
Structure forms not only in homogeneous media (Jeans instability) but also in flattened disks (Toomre Instability).

In these systems, the mostly 2D configuration and the presence of differential rotation (shear) will lead to different stability properties.



Toomre Instability

In addition to sound waves removing compressibility, in the presence of differential rotation, a collapsing region will also now have to contend with shear.



The collapse has to be faster than the timescale in which the shear is advecting the blobs apart.

Toomre Instability

To collapse, a region must contract gravitationally faster than acoustics can remove the compression and differential rotation can shear it away.

Timescale for acoustics: sound-crossing time

$$t_c = \lambda/c_s$$

Timescale for gravity: free-fall time

$$t_g = 1/(G\rho)^{1/2}$$

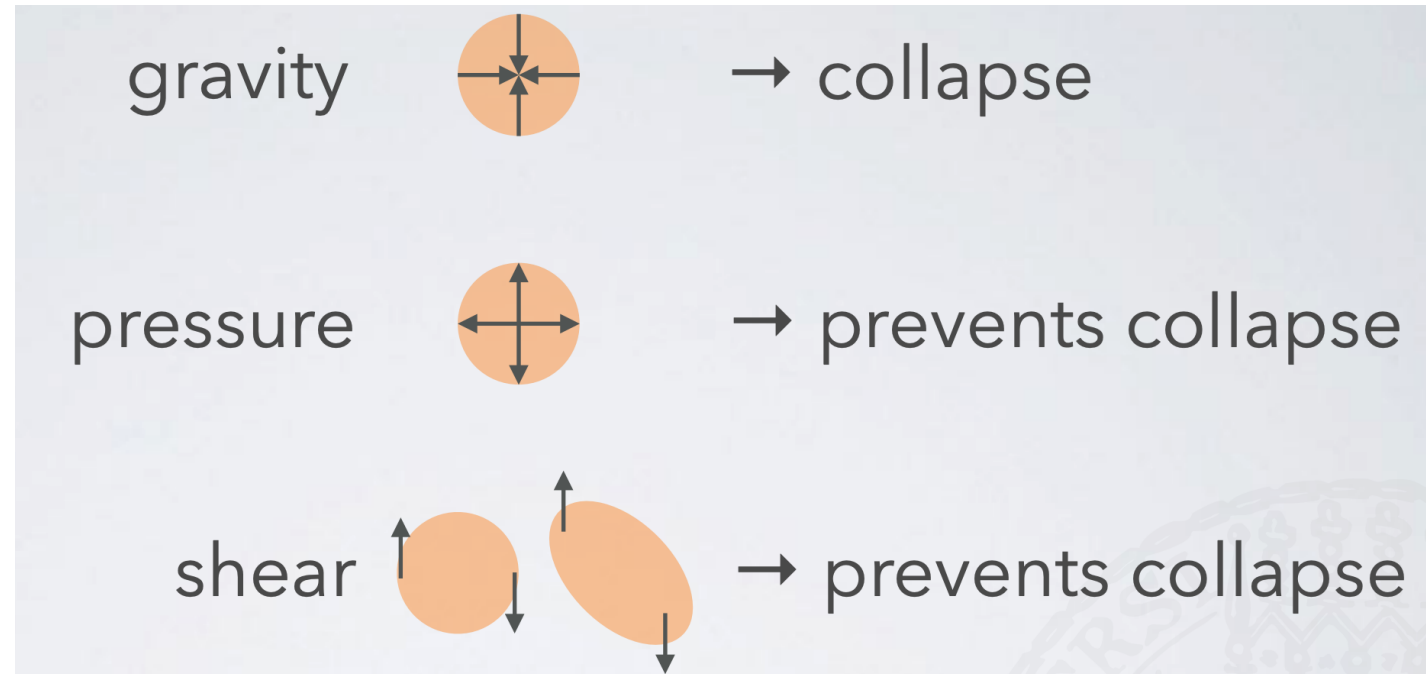
Timescale for shear: orbital time

$$t_s = 1/\Omega$$

Constructing a dimensionless number from
"Gravity vs Shear and Pressure"

$$Q \sim t_g^2/(t_c t_s)$$

$$\sim c_s \Omega/(G\rho)$$



If $Q > 1$, gravity is slower than the combined action of pressure and shear, and the system is stable. If $Q < 1$, gravity is faster, and the system collapses. This analysis is off from the actual Q from rigorous linear analysis by a factor π^{-1} , and the fact that shear stability is given by the epicyclic frequency κ instead.

Toomre Instability

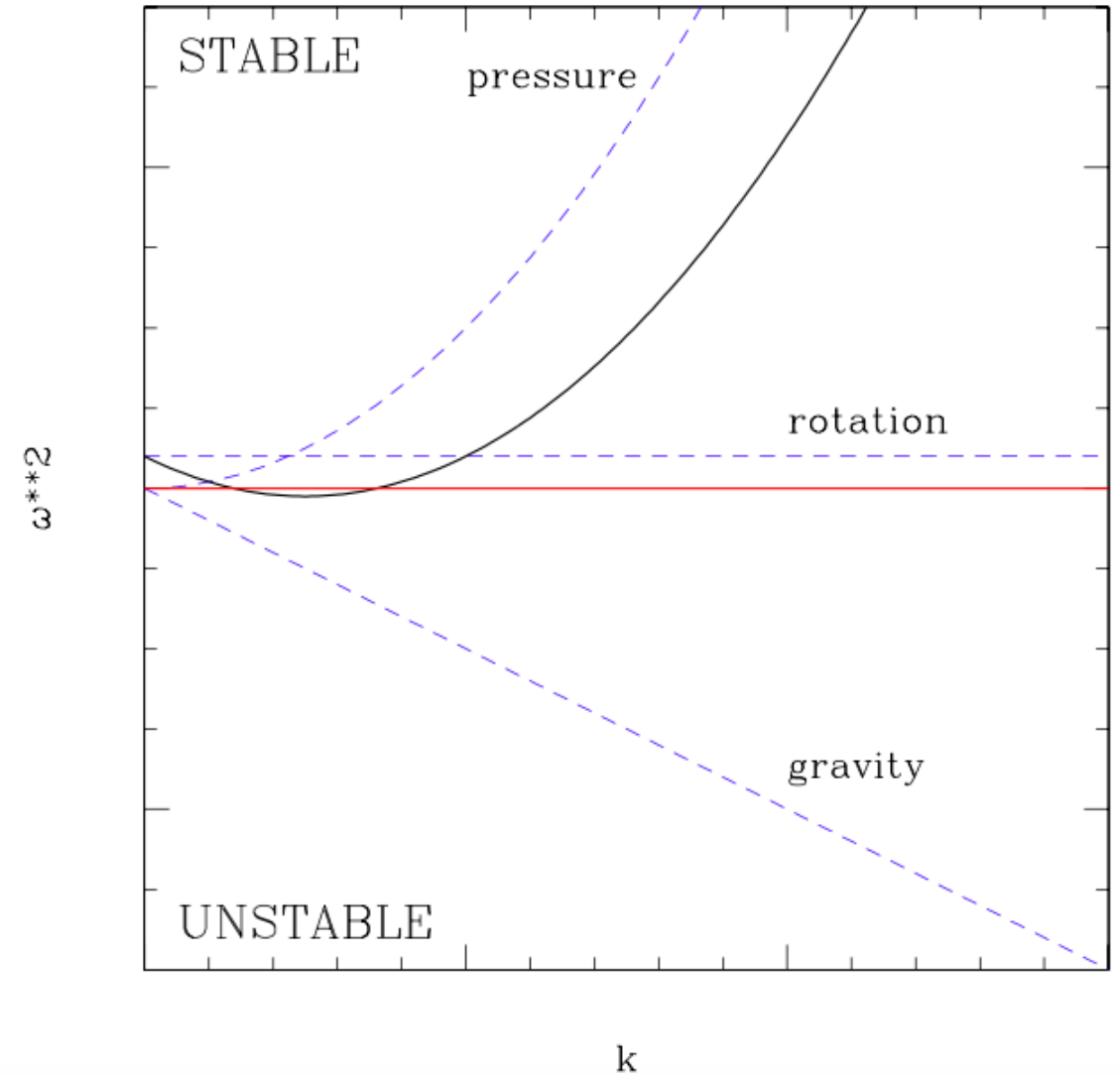
Dispersion relation

$$\omega^2 = \kappa^2 - 2\pi G \Sigma |k| + c_s^2 k^2$$

κ – epicyclic frequency

k – wavenumber

Gravity pushes ω to imaginary values. Pressure and rotation push it to real values.



Toomre Instability: Self-regulation

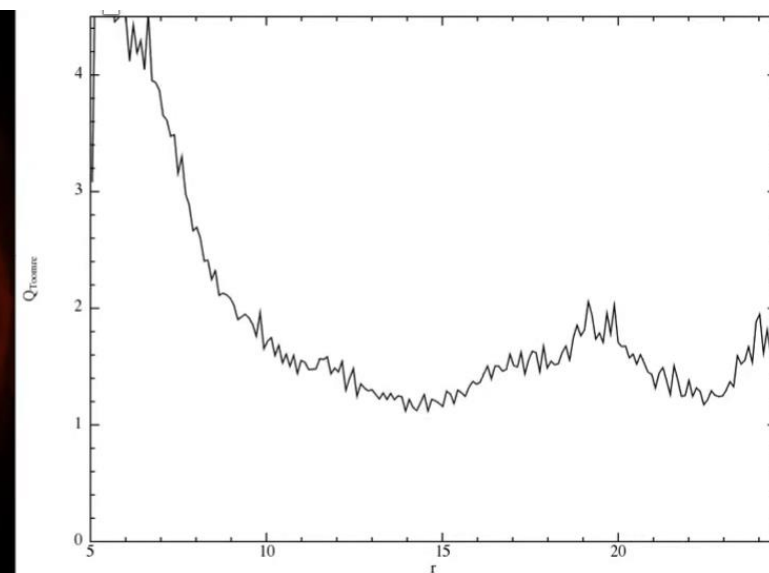
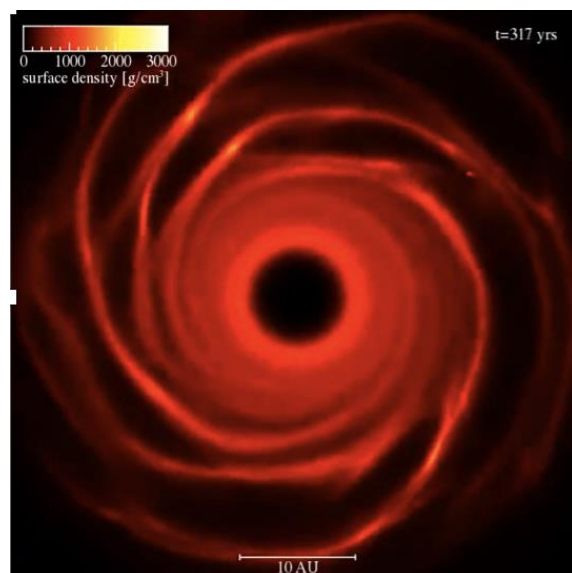
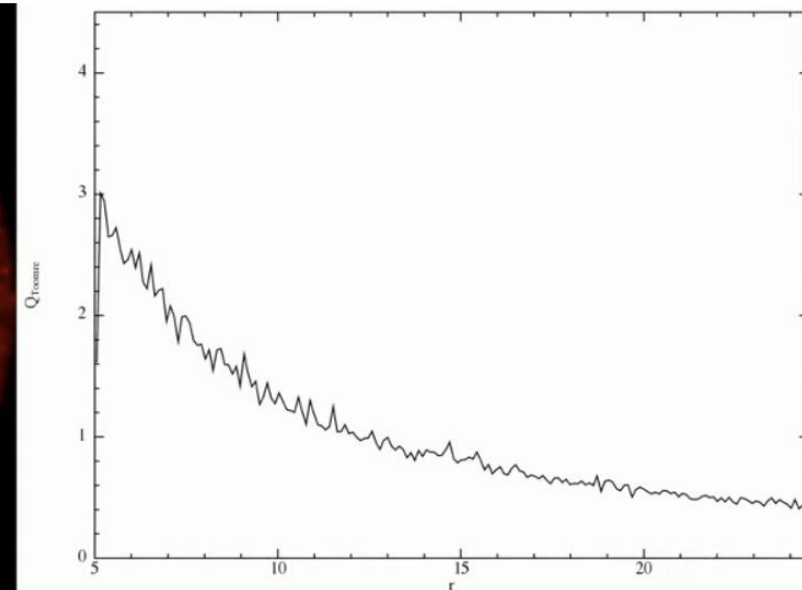
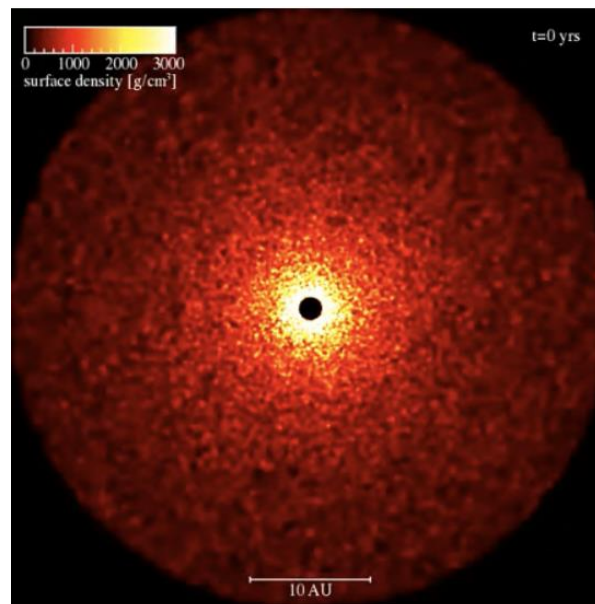
$$Q = c_s \kappa / (\pi G \rho)$$

An adiabatic system evolves toward $Q \sim 1$.

- If the system compresses (lowering Q), it also heats up (increasing Q).
- If the system expands (increasing Q), it also cools down (decreasing Q).

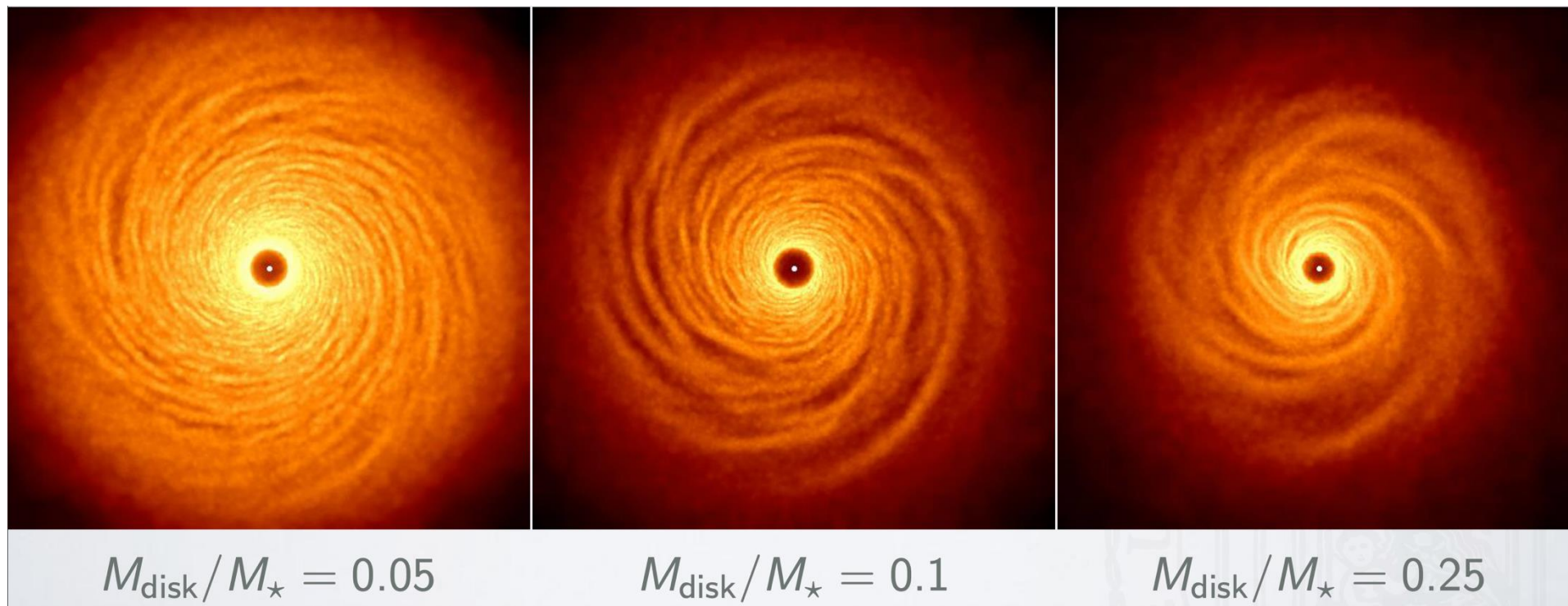
The process is self-regulated, leading to *gravitational turbulence* and *spiral arms*.

Cooling has to be fast to allow for collapse.



Toomre Instability: Self-regulation

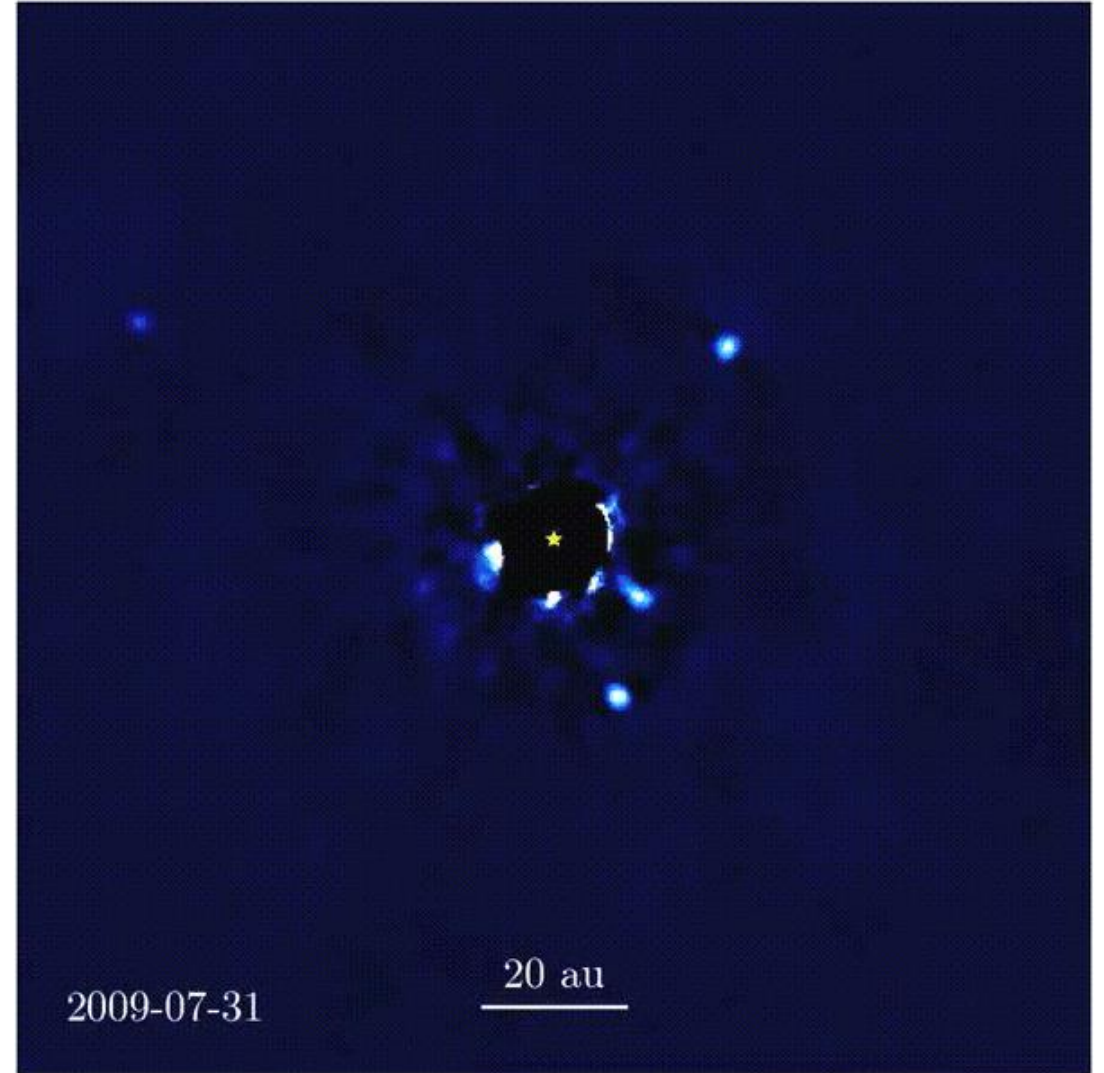
Toomre instability is slow cooling disks as the disk mass is increased.
The system passes from gravitational turbulence (low mass) to spiral arms (higher mass).
If the mass is further increased, collapse is inevitable.



Toomre Instability and Giant Planets

Toomre instability might be responsible for the formation of giant planets in wide orbits.

The process is also called simply “gravitational instability” or “disk instability” in the planet formation literature.



Typical Toomre Q values



Galactic disks

$$Q \sim 1$$



Starburst galaxies

$$Q < 1$$